MILITARY SPECIFICATION

MODULE ASSEMBLY, PALLET TYPE, STORAGE

This specification is approved for use within U.S. Marine Corps, Department of the Navy, and is available for use by all departments and agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the requirements for manufacture and acceptance of the Module Assembly, Pallet Type, Storage. This item is the only one of its type.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

V-T-295	-	Thread, Nylon.
QQ-A-200/8	-	Aluminum Alloy 6061, Bar, Rod, Shapes, Tube and Wire, Extruded.
QQ-A-200/9	-	Aluminum Alloy 6063, Bar, Rod, Shapes, Tube and Wire, Extruded.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commandant of the Marine Corps, Headquarters Marine Corps (Code LMA-1), Washington, DC 20380 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 7125

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T23:09Z
Check the source to verify that this is the current version before use.

QQ-A-225/8	-	Aluminum Alloy 6061, Bar, Rod, Wire and Special Shapes, Rolled, Drawn or Cold Finished.
QQ-A-250/8	-	Aluminum Alloy 5052, Plate and Sheet.
QQ-S-781	-	Strapping, Steel and Seals.
PPP-B-601	-	Boxes, Wood, Cleated Plywood.
PPP-B-621	-	Box, Wood, Nailed and Lock-corner.

MILITARY

MIL-P-116	-	Preservation, Methods of.
MIL-B-117	-	Bag, Sleeve and Tubing - Interior Packaging
MIL-T-704	-	Treatment and Painting of Material.
MIL-W-4088	-	Webbing, Textile, Woven Nylon.
MIL-C-5541	-	Chemical Conversion Coatings on Aluminum Alloys.
MIL-E-6060	-	Envelope, Packaging, Water Vapor Proof, Flexible.
MIL-G-12803	-	Gasket Material, Non-Metalic.
MIL-P-19834	-	Plate, Identification, Metal Foil, Adhesive Back.
MIL-C-22750	-	Coating, Epoxy Polyamide.
MIL-W-45205	-	Welding, Gas Metal-Arc and Gas Tungsten-Arc, Aluminum Alloys, Readily Weldable for Structures, Excluding Armor.
MIL-P-53022	-	Primer, Epoxy Coating, Corrosion Inhibiting, Lead and Chromate Free.

STANDARDS

FEDERAL

FED-STD-595	-	Federal Standard Colors.

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	-	Marking for Shipment and Storage.
MIL-STD-130	-	Identification Marking of U.S. Military Property.
MIL-STD-2073-1A	-	DoD Material Procedures for Development and Application of Packaging Requirements.

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T23:09Z
Check the source to verify that this is the current version before use.

DRAWINGS

U.S. Marine Corps Logistics Base, Albany, GA
82A5048A1900 - Module Assembly, Pallet Type, Storage
82A5050A0100 - Rail Assembly and Installation, Mounting, Floor
82A5050A0110 - Peg Assembly, Modular Alignment
82A5050A0132 - Bar, Lock
82A5050A0138 - Shaft, Latch
82A5050A0140 - Head, Latch
82A5050A0142 - Handle, Latch
82A5050A0144 - Latch
82A5050A0150 - Rail Assembly and Installation, Mounting, Wall
82A5050A0160 - Post Assembly, Mounting

(Copies of specifications, standards, handbooks, drawings, publications, and other government documents required by contractors in connection with specific procurement functions should be obtained from the contracting activity or as directed by the contracting activity).

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

AMERICAN NATIONAL STANDARDS INSTITUTE

ANSI/AWS A2.4-79 - Welding and Nondestructive Testing, Symbols.
ANSI/AWS A3.0-85 - Welding Terms and Definitions.

(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T23:09Z
Check the source to verify that this is the current version before use.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.3 and 6.4).

3.2 Materials. The materials and components shall be certified to conform to the applicable specification and standards as listed on the drawing package 82A5048A1900 or required herein. The materials used may include reclaimed materials provided such materials will not jeopardize its intended use and performance. The reclaimed materials shall have been reprocessed, remanufactured, or recycled in a manner which will restore them to the same chemical composition and physical properties as the materials originally selected for use as specified herein.

3.3 Design. Module Assembly, Pallet Type, Storage, shall be of the design specified on drawing package 82A5048A1900 (see 2.1.2).

3.4 Construction. Module Assembly, Pallet Type, Storage, shall be of the construction and physical dimensions specified on drawing package 82A5048A1900 (see 2.1.2).

3.4.1 Frame assembly. The frame assembly shall be fabricated from Rod, Aluminum Alloy, 6061-T6, QQ-A-225/8; Aluminum Alloy 6063-T6, QQ-A-200/9; Angle, Aluminum Alloy, 6061-T6, QQ-A-200/8; and shall be square in all directions within .010 inch per foot (see 4.4.2). All welds shall be in accordance with ANSI/AWS A3.0-85, A2.4-79 and MIL-W-45205, Class B (see 4.4.4). The frame assembly shall conform to specifications to such a degree of accuracy ensuring it will align to the peg and pin assemblies and lock using the swing latches as illustrated in the footprint drawing, Figure 1 (see 4.4.3).

3.4.2 Pallet assembly. The pallet assembly shall be fabricated from Sheet, Aluminum Alloy 5052-H32, .19 inch thick, QQ-A-250/8. The assembly shall be mounted to the frame using 6 ring and stud assemblies and 2 bolts (see 4.4.5).

3.4.3 Strap assembly, tie down. The strap assembly, tie down shall be fabricated from Strap, Webbing, MIL-W-4088, Type II, Class I, 2500 lbs minimum strength, color Gray; Strap, Webbing MIL-W-4088, Type XXVII, Class I, 6500 lbs minimum strength, color Gray; sewn with Thread per Fed Spec V-T-295, Type II, Class A, Type B, color Black per FED-STD-595. Attaching hardware will consist of Rotating End Fitting; Plate, Threading; Hook, Flat; and Fastener, Flat Hook. Attaching hardware shall be Cold Rolled Steel, 300 lbs minimum strength with HD Zinc Plate Bronze Chromate Finish. The strap assembly shall be fabricated to have a minimum strength of 300 lbs (see 4.4.6).

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T23:09Z
Check the source to verify that this is the current version before use.

3.4.4 Strap assembly. The strap assembly shall be made from Strap, Webbing Type XXVII, Class 1 and Strap, Webbing Type II, Class 1. The webbing color shall be Forest Green and all ends shall be fused to provide a smooth edge with the cut ends all fused together (see 4.4.7).

3.4.5 Bracket mounting. The bracket, mounting, shall be fabricated of Angle, Aluminum Alloy, 6061-T6, QQ-A-200/8 and shall be bolted to the frame in 4 places (see 4.4.7). The bracket, mounting, shall be bolted to the frame to such a degree of accuracy ensuring it will align to the footprint as described in Figure 1 (see 4.4.8).

3.4.6 Decal. The decal shall be fabricated from .003 inch thick aluminum alloy foil, photographically sensitized, adhesive back in accordance with MIL-P-19834, Type I. Requirements and methods of marking the decal shall be in accordance with MIL-STD-130. Unless otherwise specified all characters shall be .093 inch high on finished decal, color shall be in accordance with MIL-P-19834, Style III (see 4.4.9). It shall be attached to the frame center horizontal member 5 inches from the left edge.

3.5 Finish.

3.5.1 Cleaning. Prior to painting metal surfaces shall be cleaned and treated per MIL-T-704, Type G, and MIL-C-5541, Class 3.

3.5.2 Priming. After metal surfaces have been cleaned as required they shall be primed per MIL-P-53022, with a minimum dry film thickness of .001 inch in one coat.

3.5.3 Painting. Final finish painting shall have a minimum dry coat film thickness of .0018 inch in two coats per MIL-C-22750, Type II, Color No. 24533 per FED-STD-595 (see 4.4.12).

3.6 Interchangeability. All parts having the same part number shall be functionally and dimensionally interchangeable. Interchangeable parts are defined as two or more like parts possessing such functional and physical characteristics as to be equivalent in performance and durability, and capable of being exchanged one for the other without alteration of the parts themselves or adjoining parts and without selection for fit or performance.

3.7 Workmanship. The Module Assembly, Pallet Type, Storage shall meet all design, fabrication, assembly and finish requirements of this specification. Particular attention shall be given to; removal of burrs and sharp edges; accuracy of dimensions; thoroughness of welding; painting; alignment; tightness of screws, bolts, and rivets. Failure to ensure these measures have been taken will be considered adequate basis for rejection of items being of inferior quality for the purpose intended.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T23:09Z
Check the source to verify that this is the current version before use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of section 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Test equipment and inspection facilities. When specified (see 6.2) the contractor shall insure that test and inspection facilities of sufficient accuracy, quality and quantity are established and maintained to permit performance of required inspections.

4.2 Classification of inspections. The inspections specified herein are classified as follows:

a. First article inspection. First article inspection consists of examinations and tests performed on samples which are representative of the production item after award of a contract to determine that the production item conforms to the requirements of this specification (see 4.3).

b. Quality conformance inspection. Quality conformance inspection consists of examinations and tests performed on individual products to determine conformance of the products with the requirements set forth in this specification (see 4.4).

c. Visual inspection. Visual inspection consists of examinations to determine conformance with the requirements set forth in this specification (see 4.5).

4.3 First article inspection. The first article inspection of the Module Assembly, Pallet Type Storage shall consist of examination and tests for all the requirements of this specification.

Source: https://assist.dla.mil -- Downloaded: 20260-20T23:09Z
Check the source to verify that this is the current version before use.

4.3.1 First article sample. Unless otherwise specified as soon as practical after the award of the contract, the contractor shall submit quantity(s) specified in the contract for Module Assembly, Pallet Type, Storage. The modules shall be representative of the construction, workmanship, components, and materials to be used during production. Approval of the first article inspection samples or the waiving of the first article inspection does not preclude the requirements for performing the quality conformance inspection. The first article inspection samples shall be furnished to the Government as directed by the Contracting Officer (see 6.2).

4.3.2 Sample disposition. Upon completion of the first article inspection, sample(s) will be disposed of in accordance with Government shipping instructions.

4.3.3 First article inspection routine. First article inspection shall consist of those items listed in Table I.

4.4 Quality conformance inspection. Quality conformance inspections shall be conducted in accordance with MIL-STD-105 and consist of those items listed in Table I.

TABLE I. Tests and Inspections

Requirement	Requirement Paragraph	Test Paragraph
Condition and certification of materials	3.4.1, 3.4.2 3.4.5	4.4.1
Dimensional check of frame assembly	3.4.1, 3.4.4	4.4.2
Verification Frame Assembly fits footprint of Figure 1	3.4.1	4.4.3
Welding	3.4.1	4.4.4
Pallet assembly	3.4.2	4.4.5
Strap assembly, tie down	3.4.3	4.4.6
Strap assembly	3.4.4	4.4.7
Bracket, mounting	3.4.5	4.4.8
Decal	3.4.6	4.4.9
Cleaning of aluminum	3.5.1	4.4.10
Priming	3.5.2	4.4.11

Source: https://assist.dla.mil -- Downloaded: 202610-20T23:09Z
Check the source to verify that this is the current version before use.

TABLE I. Tests and Inspections (Cont'd)

Requirement	Requirement Paragraph	Test Paragraph
Marking	5.3	4.4.14
Hardware	3.4.2, 3.4.5	4.4.15
Painting	3.5.3	4.4.12
Packaging	5.2	4.4.13

4.4.1 Condition and certification of materials. Material used in construction of the Module Assembly, Pallet Type, Storage, (sheet, roll and bar aluminum) shall be certified to be of the specified material that has no etching, scaling or pitting. Fabricated pieces shall not have sharp edges or burrs. Material shall not be cracked, malformed or split.

4.4.2 Frame dimensions. The frame assembly shall be measured and shall be square in all directions to within .010 inch per foot.

4.4.3 Footprint fit. The frame assembly shall be inspected for alignment to the peg and pin assemblies and lock, using the swing latches as illustrated in the footprint drawing Figure 1.

4.4.4 Welding. Welding areas shall be inspected for cleanliness and be free of dirt, sand, scale and oil prior to welding. All welds shall be inspected for compliance with ANSI/AWS A3.0-85, A2.4-79 and MIL-W-45205 Class B.

4.4.5 Pallet assembly. The pallet assembly shall be inspected for attachment to the frame using 6 ring stud assemblies and 2 bolts.

4.4.6 Strap assembly, tie down. The strap assembly tie down shall be tested for a minimum strength of 300 lbs.

4.4.7 Strap assembly. The strap assembly shall be inspected for compliance with 3.4.4.

4.4.8 Bracket mounting. The bracket mounting shall be inspected for secure attachment to the frame to such a degree of accuracy to align with the footprint as described in Figure 1.

4.4.9 Decal. The decal shall be inspected for attachment to the Module Assembly, Pallet Type, Storage, as specified on the drawings and color to be in accordance with MIL-P-19834, Style III.

Source: https://assist.dla.mil -- Downloaded: 20260-20T23:09Z
Check the source to verify that this is the current version before use.

4.4.10 Cleaning of aluminum. Cleaning of aluminum surfaces prior to painting shall be inspected for compliance with MIL-T-704, Type G, and MIL-C-5541, Class 3.

4.4.11 Priming. Priming shall be inspected for compliance with MIL-P-53022.

4.4.12 Painting. Painting of the Module Assembly, Pallet Type, Storage, shall be inspected for compliance with MIL-C-22750, Type II, Color No. 24533 per FED-STD-595 with final finish coat shall have a minimum dry coat thickness of .0018 inch in two coats. Coatings shall be inspected to have a level adherent continuous and uniform film without runs, wrinkles, streaks, or areas of no film.

4.4.13 Packaging. Packaging shall be inspected for compliance with MIL-STD-2073-1A.

4.4.14 Marking. Marking on the shipping container shall be inspected for compliance with MIL-STD-129.

4.4.15 Hardware. All hardware shall be inspected to be in place and securely fastened.

4.5 Visual inspection. The Module Assembly, Pallet Type, Storage, shall be examined for defects listed in Table II. The defects shall be classified as specified in Table II.

TABLE II. Classification of Defects for Visual Inspection

Defect	Classification	
	Major	Minor
Frame or components misaligned	X	
Welds cracked or incomplete	X	
Metal components dented, misshaped, distorted or otherwise defective	X	
Sharp burrs or metal slivers	X	
Hardware missing	X 1/	
Hardware not securely fastened		X
Decal mounted incorrectly		X

Source: https://assist.dla.mil -- Downloaded: 20260-20T23:09Z
Check the source to verify that this is the current version before use.

1/ Major when it affects the strength of attachment; otherwise, it is to be classified as a minor defect.

5. PACKAGING

5.1 Preservation. Preservation shall be level A in accordance with MIL-STD-2073-1A.

5.1.1 Unit packs. Each module shall be individually unit packed in accordance with submethod 1A-16 (Floating Bag Sealed), of MIL-P-116. Modules preserved, wrapped, and cushioned as required in paragraph 3.6.3 of MIL-P-116, shall be enclosed in a sealed bag conforming to MIL-B-117, type I or II; class E, F, or G, Style 1 or 2. Bags in accordance with MIL-E-6060 shall be used for sizes larger than can be made in accordance with MIL-B-117. The bag shall be provided with gasketed holes to permit the installation of the barrier over and around the fasteners used to secure the item(s) to interior supports or to one face of the container. Gaskets shall be attached to the barrier with adhesive. Unless otherwise specified, material for gaskets shall conform to MIL-G-12803 (see paragraph 6.2.1 (g)). Material for gaskets shall not contain asbestos.

5.2 Packing. Packing shall be level B in accordance with MIL-STD-2073-1A.

5.2.1 Level B. Modules, preserved as specified in 5.1, shall be packed in wood containers conforming to PPP-B-601, overseas type of PPP-B-621, class 2. Closure and strapping shall be in accordance with the applicable container specification except that metal strapping shall conform to QQ-S-781, type I, finish A.

5.3 Marking. In addition to any special marking required by the contract or order, shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The Module Assembly, Pallet Type, Storage and other modules will be utilized in furnishing an Electronic Maintenance Complex (EMC) which will support a given mission. The number and type of modules installed in any EMC or group of EMCs, at any given time, will depend directly on the user's immediate needs. Its function will afford personnel a dedicated fixture to stow large coils of power cable, boxes, frames, assorted pieces of hardware or support equipment.

6.2 Ordering data.

6.2.1 Acquisition requirements. Procurement documents shall specify the following:

Source: https://assist.dla.mil -- Downloaded: 20260-20T23:09Z
Check the source to verify that this is the current version before use.

a. Title, number, and date of this specification.

b. Quantity desired.

c. When first article testing is required (see 3.1).

d. When test equipment and inspection facilities are required (see 4.1.2).

e. When quality conformance testing is required (see 4.4).

f. Name and address of first article inspection facility.

g. Level of preservation, packaging, and packing required (see section 5).

h. Whether any special markings are required.

6.3 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL) the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of DOD FAR Supplement, Part 27, Sub-Part 27.410-6 are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraphs.

Data Requirements	Applicable DID Number	Options
First Article Inspection Procedure	DI-T-4901	-------
First Article Inspection Report	DI-T-4902	-------
Production/Acceptance Inspection Procedures	DI-T-4903	-------
Production Inspection Report	DI-T-4904	-------

(Copies of DID's required by contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.4 First article. The first article is to be examined and tested for approval at the contractor's plant or at an independent commercial testing laboratory acceptable to the procuring activity. First article tests shall be witnessed by a Government Representative of the procuring activity. Approval of first articles shall be by the procurement contracting officer.

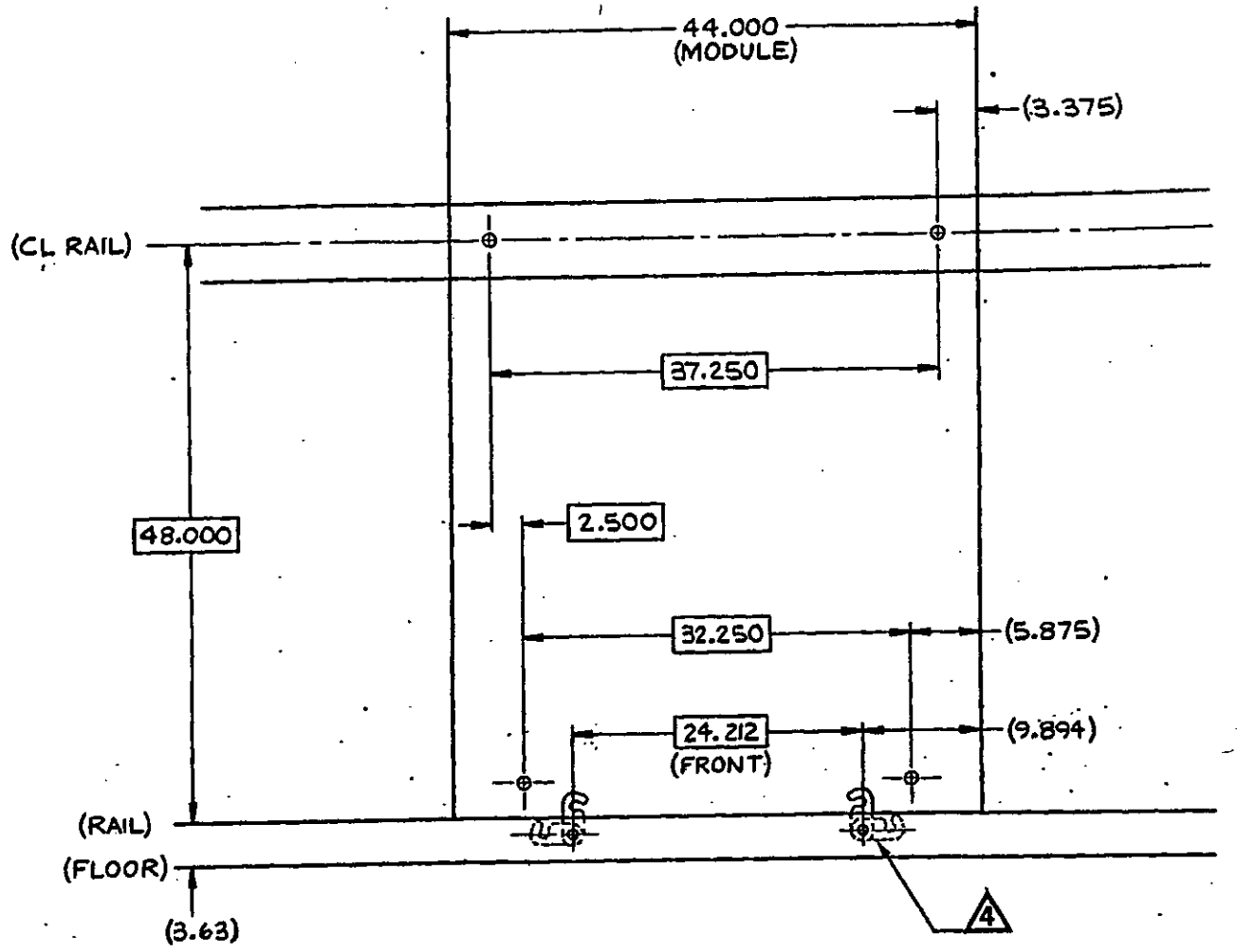
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T23:09Z
Check the source to verify that this is the current version before use.

6.5 Subject term (key word) listing. The following subject term (key word) listing is furnished for retrieval searches.

Assembly, frame
Assembly, module, pallet type, storage
Assembly, strap
Assembly, strap, tie down

Preparing activity:
Navy-MC
Project No. 7125-N122

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T23:09Z
Check the source to verify that this is the current version before use.



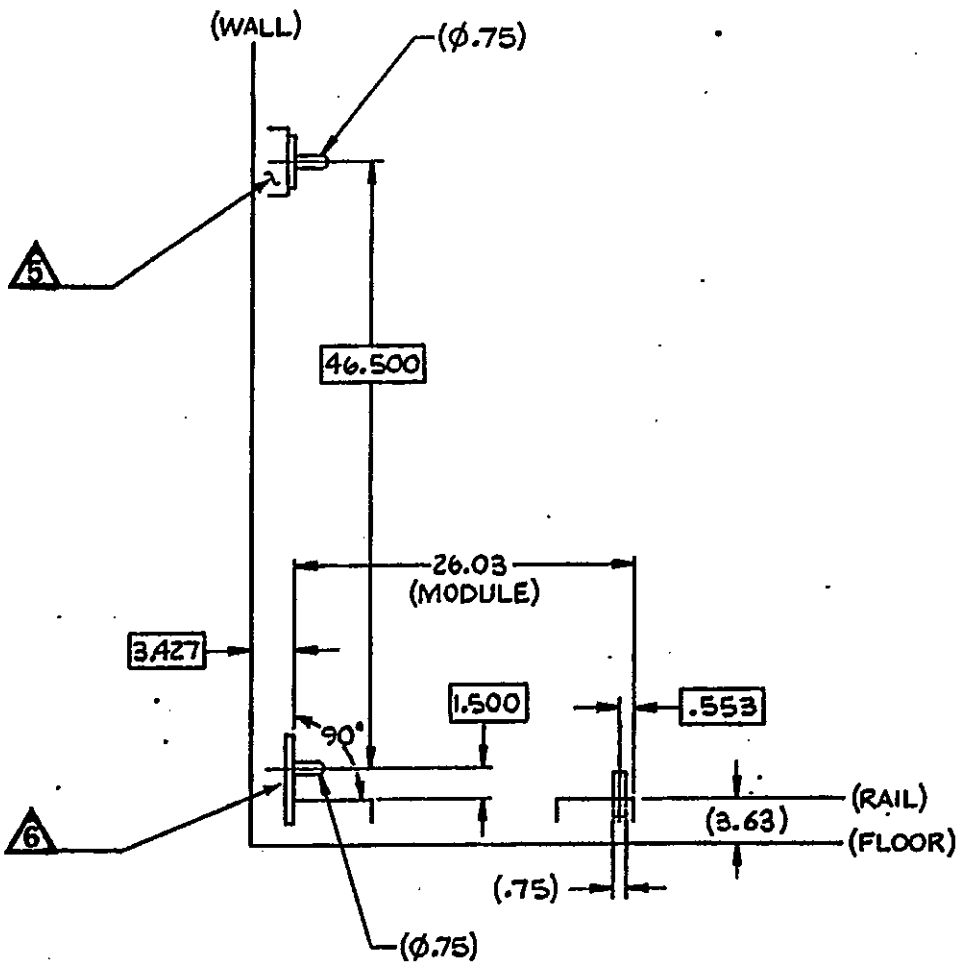
FRONT VIEW

NOTES:
1. Dimensions are in inches.
2. Unless otherwise specified, tolerance is .XX ±.03, .XXX ±.010.
3. Perpendicularity equals ±1°.

4 REF DWGS: 82A5050A0100 (RAIL ASSEMBLY AND INSTALLATION, MOUNTING, FLOOR)
82A5050A0132 (BAR, LOCK)
82A5050A0138 (SHAFT, LATCH)
82A5050A0140 (HEAD, LATCH)
82A5050A0142 (HANDLE, LATCH)
82A5050A0144 (LATCH)

Figure 1. (Page 1 of 2)
Module Assembly Footprint

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T23:09Z
Check the source to verify that this is the current version before use.



SIDE VIEW

5 REF DWGS: 82A5050A0150 (RAIL ASSEMBLY AND INSTALLATION, MOUNTING, WALL)
82A5050A0160 (POST ASSEMBLY, MOUNTING)

6 REF DWGS: 82A5050A0100 (RAIL ASSEMBLY AND INSTALLATION, MOUNTING, FLOOR)
82A5050A0110 (PEG ASSEMBLY, MODULAR ALIGNMENT)

Figure 1. (Page 2 of 2)
Module Assembly Footprint

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T23:09Z
Check the source to verify that this is the current version before use.